Aussie Primary Academy

Year 3 Maths — 2D Shapes: Compare and Classify

Name: ______ Date: ______

ACARA: AC9M3SP01

Learning Intention: I can compare and classify 2D shapes using their sides and vertices.

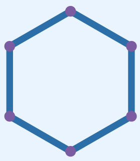
Success Criteria — I can:

- Count and record the sides and vertices of a 2D shape.
- Name common and less familiar 2D shapes, including triangles, quadrilaterals, pentagons, hexagons and octagons.
- Classify a shape using more than one feature.

Strategy Box

To classify a 2D shape: **1)** Count the sides. **2)** Count the vertices (corners). **3)** Check if all the sides and angles look equal — this tells you if it is regular or irregular. **4)** Use the side count to name the shape.

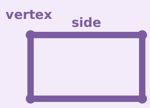
Worked Example



1. Count the sides → this shape has **6 straight sides**.
2. Count the vertices → there are **6 vertices** (corners) where sides meet.
3. All sides and angles look equal, so it is a **regular** shape.
4. A shape with 6 sides is called a **hexagon**.

Review: Sides and Vertices

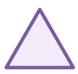
A **side** is a straight line that makes part of the shape's outline. A **vertex** (plural: vertices) is the point where two sides meet.



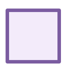
Quick Facts

Sides	Shape name
3	Triangle
4	Quadrilateral
5	Pentagon
6	Hexagon
8	Octagon

Common and Less Familiar 2D Shapes

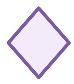


Triangle



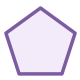
Square

Rectangle

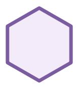


Rhombus

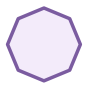
Trapezium



Pentagon



Hexagon



Octagon

Rectangle, rhombus and trapezium are all **quadrilaterals** — they have 4 sides, just like a square.

Parent Tip

Go on a shape hunt around your home! Ask your child to find something shaped like a triangle, a rectangle and a hexagon (a stop sign or nut/bolt is octagon-shaped). Ask them to count the sides together.

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Build Skills — Compare, count and classify 2D shapes.

Q1 — Count and Record

Look at each shape. Count the sides and vertices, then write the shape's name.

Sides: ___ Vertices: ___

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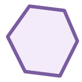
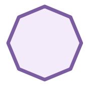
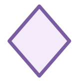


Sides: ___ Vertices: ___



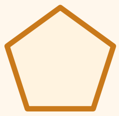
Sides: ___ Vertices: ___



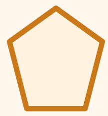
Sides: ___ Vertices: ___

Q2 — Compare Two Shapes

Look carefully at Shape A and Shape B. Answer the questions below.



Shape A



Shape B

a) How many sides does each shape have? A: ___ B: ___
b) Are Shape A and Shape B both the same type of shape? ______
c) What is one difference you can see between Shape A and Shape B? ________________________

Q3 — Classify Using More Than One Feature

Circle all the shapes that have **4 sides AND all sides look equal**.

Tip: a shape can look tilted and still be the correct shape — check the sides, not just how it is sitting on the page.

Q4 — Sort the Quadrilaterals

Write each quadrilateral's letter in the correct box below.

A B C D E

All sides equal length (square / rhombus)	Opposite sides equal, no sides all equal (rectangle / parallelogram)	Only one pair of parallel sides (trapezium)

Watch Out!

A square that is tilted (rotated) is still a square — rotating a shape does not change its name, its number of sides or its number of vertices. Do not rely on colour alone to identify a shape — always check the sides and vertices.

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Apply & Reasoning — Use what you know to explain your thinking.

1 **Reasoning:** Zara says, "All rectangles are squares." Is she correct? Explain your thinking using the words sides and equal.

2 **Reasoning:** A hexagon and an octagon both have equal sides and equal angles. How can you tell them apart?

3 **Open-ended:** Draw and name a 2D shape with **more than 4 sides**. Label its sides and vertices with numbers.

Quick Check — Exit Ticket

Write the name of a 2D shape with **6 sides** and draw one small example of it in the space below.

Answer & Teacher Support

Answer Key — Pages 1 & 2

Worked example (p1): 6 sides, 6 vertices, regular hexagon.

Q1 (p2): Triangle 3/3; Rectangle 4/4; Irregular hexagon 6/6; Octagon 8/8; Rhombus 4/4 (sides/vertices).

Q2 (p2): A and B both pentagons, 5 sides each; accept any visible difference named (e.g. size/regularity).

Q3 (p2): Circle square, rectangle, rhombus and rotated square — all 4 equal sides. Not trapezium or triangle.

Q4 (p2): Equal sides: A, C. Opposite sides equal: B, E. One pair parallel: D.

Answer Key — Page 3

Q1: No — a rectangle has 4 sides but only a square has all sides equal. Every square is a rectangle; not every rectangle is a square.

Q2: Count sides/vertices — hexagon = 6, octagon = 8.

Q3 (acceptable-answer guide): Accept any shape with 5+ straight sides where labelled sides = labelled vertices and the name matches the side count.

Exit ticket: Any correctly named hexagon with a 6-sided drawing.

Teacher Checklist

- ☐ Counts sides/vertices accurately.
- ☐ Names triangles, quadrilaterals, pentagons, hexagons, octagons.
- ☐ Classifies using more than one feature.
- ☐ Recognises a rotated shape is unchanged.
- ☐ Sorts quadrilaterals appropriately.
- ☐ Justifies reasoning with maths vocabulary.

Teacher Tip

Covers the **2D** half of AC9M3SP01 — complements (does not repeat) the "3D Objects" roll/stack/slide worksheet. Watch for the common misconception that a rotated square becomes a "diamond" shape and stops being a square.